Intelligent Electricity Monitoring, Theft Detection, and Consumption Forecasting: Methods, Challenges, and the VoltIQ Hardware-Integrated Framework

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Abstract- ***Electricity theft and inefficient consumption tracking cause significant revenue losses for utility providers and make accurate demand forecasting difficult. Conventional metering offers little real-time anomaly detection and no predictive insight into usage patterns. This review organizes the literature around five complementary directions reflected in the supplied studies: low-cost IoT sensing hardware, supervised electricity theft classification, class-imbalance handling with SMOTE, unsupervised anomaly detection with Isolation Forest, and consumption forecasting with Prophet and XGBoost. The reviewed studies show a progression from basic sensing and billing estimation to tree-ensemble theft classifiers, hybrid anomaly detectors with interactive dashboards, and hybrid forecasting models. We synthesize these developments into a reference-guided framework, VoltIQ, that separates sensing, data logging, forecasting, anomaly detection, theft classification, dashboard alerting, and relay-based load control. The paper also discusses class imbalance, false positives, generalization across consumer types, sensor accuracy, and edge-hardware constraints. Because VoltIQ is under development and the present work is a literature-based synthesis rather than a new benchmark experiment, reported performance values are identified as findings from the cited studies rather than results of the proposed framework.***

Keywords- ***Electricity theft detection; smart metering; ESP32; anomaly detection; Isolation Forest; XGBoost; SMOTE; load forecasting; Prophet; non-technical losses; SDG 7***

CHAPTER 1: INTRODUCTION

Electricity theft is a primary concern for utility providers because it leads to substantial financial losses [1]. It is a major contributor to non-technical losses in power systems [2,3], and traditional detection by manual inspection is inadequate against advanced techniques such as meter tampering and cyberattacks on smart grids [3]. At the same time, utilities need reliable demand forecasts to plan generation and distribution, and consumers need visibility into how they use energy. Conventional meters provide neither real-time anomaly detection nor predictive insight. Improving this situation supports Sustainable Development Goal 7, Affordable and Clean Energy.

A second difficulty is the nature of the data. Theft is rare compared with normal consumption, so real-world datasets are heavily imbalanced; the reviewed studies address this either by generating artificial theft patterns [1] or by oversampling with SMOTE [2,3]. A third difficulty is the measurement itself: low-cost sensing hardware introduces calibration error that propagates into every downstream model [7]. A reliable system must therefore treat sensing, data quality, learning, and decision-making as one pipeline rather than separate problems.

The technical evolution of the field can be viewed as a sequence of increasingly capable layers. Low-cost ESP32-based monitors made real-time voltage and current acquisition affordable [7]. Supervised tree-ensemble classifiers, especially XGBoost, became the dominant approach for theft detection [1,2,3]. Unsupervised methods such as Isolation Forest label abnormal consumption without requiring labeled theft data [4,5], and hybrid forecasting models combine Prophet with XGBoost for load prediction [6].

This paper is a literature-guided review of that evolution, written for the VoltIQ project. Rather than presenting unsupported experimental claims, it compares seven studies, extracts the design principles they establish, and derives a unified conceptual architecture for a hardware-integrated system that measures, forecasts, detects, and responds. The central question is how a low-cost device can move from passive metering toward timely, explainable, and safe detection of theft and anomalies.

CHAPTER 2: LITERATURE REVIEW AND EVOLUTION

2.1 LOW-COST IOT SENSING HARDWARE

A common hardware pattern combines an ESP32 microcontroller, a ZMPT101B voltage sensor, and an SCT-013 current sensor. One residential prototype uses this combination with the ThingSpeak cloud platform for real-time monitoring, current-spike detection, and bill estimation [7]. The ESP32 acts as the central processing unit, the ZMPT101B measures voltage, and the SCT013-030 monitors current flow [7]. The system is reported to have acceptable accuracy at low cost, with a highest observed error difference of 12.438% [7].

The main benefit of this class of hardware is affordability and ease of integration. Its weakness is measurement error: a double-digit percentage error in the worst case shows that calibration against a reference meter is essential before readings are used for theft decisions [7]. The cited prototype also stops at monitoring and billing estimation; it does not add forecasting, theft classification, or load control [7].

2.2 SUPERVISED THEFT DETECTION

Supervised classifiers learn to separate honest from fraudulent consumption patterns. Kawoosa et al. propose an XGBoost-based model that uses consumption patterns and

region-specific variables as input features. It is reported to outperform benchmarks including random forest, support vector machine, decision tree, and AdaBoost, with a 96% detection rate and a 3% false positive rate [1]. Putro et al. integrate XGBoost with load-profile data and report 93.63% accuracy, with improved sensitivity and fewer false positives and false negatives [2].

Alfaverh et al. evaluate Random Forest, Decision Tree, XGBoost, LightGBM, CatBoost, Extra Trees, and Logistic Regression on the TDD2022 dataset, with LASSO and ReliefF for feature selection. Tree-based ensembles perform best, and consumer-specific models outperform global models [3]. Together these studies support the use of XGBoost and Random Forest as the core theft classifiers [1,2,3].

2.3 CLASS IMBALANCE AND SMOTE

Theft cases are far fewer than normal cases, so a classifier trained on raw data can achieve high accuracy while missing most theft. Kawoosa et al. rebalance their data using six artificially created theft attacks [1]. Putro et al. use SMOTE to balance normal customers against those with anomalous consumption [2], and Alfaverh et al. apply SMOTE together with feature selection, finding that Extra Trees combined with SMOTE and ReliefF gives the strongest results [3]. Imbalance handling is therefore a required part of the pipeline rather than an optional refinement [1,2,3].

2.4 ANOMALY DETECTION WITH ISOLATION FOREST

Isolation Forest is attractive because it is unsupervised and needs no labeled theft data. ELhadad et al. use it to label smart meter readings as normal or abnormal, then train Random Forest and Decision Tree models on the label sequences to predict anomaly occurrence in advance; the two supervised models perform similarly [5]. A separate study combines Z-score, local outlier factor, one-class SVM, and Isolation Forest on simulated smart meter data and flags an anomaly only when all four methods agree, which reduces false positives and outperforms single-method models [4].

These findings show two useful ideas for a theft system: unsupervised labeling can bootstrap supervised learning [5], and requiring agreement among detectors trades some sensitivity for fewer false alarms [4].

2.5 CONSUMPTION FORECASTING

Forecasting supports demand planning and also gives a baseline against which unusual usage can be judged. Zeng et al. propose a Prophet-BO-XGBoost framework: XGBoost models the nonlinear relationship between features and load, Prophet supplies a time-series view through label prediction, and Bayesian optimization tunes the XGBoost hyperparameters [6]. The motivation is limited accuracy and poor generalization of short-term load forecasting under complex nonlinear conditions [6]. This is the closest published match to the Prophet and XGBoost pairing planned for VoltIQ.

2.6 DASHBOARDS AND SYSTEM INTEGRATION

Few reviewed studies connect sensing, learning, and presentation. The anomaly-detection study implements its hybrid detector in an interactive Streamlit dashboard with real-time visualization, peak-time alerts, usage forecasts, and personalized consumption suggestions [4]. The hardware prototype sends data to a cloud platform and estimates bills [7]. Neither study includes automatic load control, which leaves a gap that VoltIQ addresses with a relay module.

CHAPTER 3: REFERENCE-GUIDED UNIFIED FRAMEWORK

The seven supplied studies suggest that a modern theft-aware meter should not choose between sensing, forecasting, anomaly detection, and classification. Instead, these components can be arranged as stages in one pipeline. The framework proposed here is conceptual and is synthesized from the reviewed literature and the VoltIQ project plan; it is not presented as an experimentally validated model.

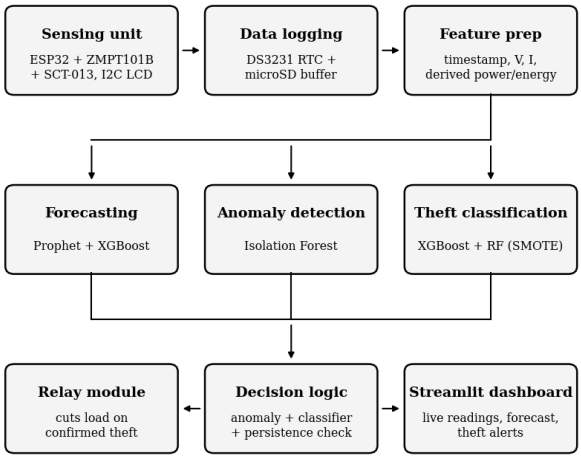


Fig. 1. Reference-guided unified pipeline synthesized from the reviewed studies.

3.1 SENSING LAYER

An ESP32 microcontroller reads a ZMPT101B voltage sensor and an SCT-013 current clamp, following the hardware pattern validated in [7]. A DS3231 real-time clock timestamps every reading, a microSD card buffers data when connectivity is lost, and an I2C LCD shows live values. Sensors should be calibrated against a reference meter because the cited prototype reports errors up to 12.438% [7].

3.2 DATA PIPELINE

Each reading is logged as timestamp, voltage, and current and exported for the machine learning models. Derived features such as power, energy, rolling statistics, and time-of-day indicators can be computed offline. A design note: with only RMS voltage and current, the computed power is an apparent-power estimate unless phase information is also captured.

3.3 FORECASTING LAYER

Prophet models trend and seasonality, while XGBoost models nonlinear dependence on engineered features, following the hybrid design of [6]. Forecast residuals provide an expected-usage band for the later decision stage. Hyperparameter search, for example Bayesian optimization as in [6], can be added once a baseline exists.

3.4 ANOMALY DETECTION LAYER

Isolation Forest scores each window of readings as normal or abnormal without needing theft labels [4,5]. Where false alarms are costly, a second detector can be added and an alert raised only on agreement, as in [4].

3.5 THEFT CLASSIFICATION LAYER

XGBoost and Random Forest classify windows as theft or non-theft [1,2,3]. SMOTE is applied to rebalance the classes [2,3], and a sound design practice is to apply it to the training split only so that test data stay untouched. Consumer-specific models should be preferred where enough data exist, since they outperform global models in [3].

3.6 DECISION, RELAY, AND FEEDBACK LAYER

The relay should not be driven by a single model output. A conceptual confirmation rule is:

$$Theft = 1 \text{ if } A(t) \geq \tau_a \text{ and } P(t) \geq \tau_p \text{ for } k \text{ consecutive windows},$$

where A(t) is the anomaly score, P(t) is the classifier's theft probability, τ_a and τ_p are thresholds, and k is a persistence length. These are design parameters; the rule describes the architecture rather than a fitted model. When the rule fires, the relay cuts the load, the event is written to the log, and the Streamlit dashboard raises an alert [4]. No reviewed study evaluates automatic load cutoff, so this stage is a design contribution that needs its own safety testing.

CHAPTER 4: COMPARATIVE ANALYSIS OF THE REVIEWED STUDIES

The comparison below summarizes the main contribution of each supplied reference. It highlights an evolution from basic sensing, to supervised theft classification, to unsupervised anomaly detection and hybrid forecasting, with no single study covering the full pipeline from sensor to load control.

TABLE I. COMPARATIVE ANALYSIS OF THE SEVEN REVIEWED STUDIES

Research Paper Title	Focus and Method	Main Contribution / Findings	Limitations / Interpretation
Using Machine Learning Ensemble Method for Detection of Energy Theft in Smart Meters [1]	XGBoost-based theft detection using consumption patterns; imbalance handled with six artificially created theft attacks.	Reports a 96% detection rate and 3% false positive rate, and outperforms random forest, SVM, decision tree, AdaBoost and other benchmarks.	Imbalance is addressed with simulated attacks, so real-world theft patterns may differ. Results are specific to the study's dataset and region.
Electricity Theft Detection Based on Load Profile Using XGBoost and SMOTE [2]	XGBoost combined with SMOTE on load-profile data to separate normal and anomalous customers.	Reports 93.63% accuracy with better sensitivity and fewer false positives and false negatives.	A short paper with a single accuracy figure in the available summary; generalization to other utilities is untested.
Electricity Theft Detection from Electricity, Gas and Water Measurements using Machine Learning [3]	Compares seven classifiers on the TDD2022 dataset, with SMOTE, LASSO and ReliefF; compares electricity-only and multi-utility inputs.	Extra Trees with SMOTE and ReliefF gives accuracy above 95% and AUC near 0.99. Consumer-specific models beat global models.	Residential profiles remain challenging, while commercial classes are near-perfect. The best model differs from the XGBoost choice in VoltIQ.
Empowering Energy Management: Anomaly Detection in Smart Meter Data for Proactive Consumption Control [4]	Hybrid detector (Z-score, LOF, one-class SVM, Isolation Forest) on simulated data, shown in a Streamlit dashboard.	Flags an anomaly only when all four methods agree; outperforms single-method models and provides alerts, forecasts and suggestions.	Uses simulated data, and requiring full agreement may miss subtle theft. No hardware or load control is included.
Anomaly Prediction in Electricity Consumption Using a Combination of Machine Learning Techniques [5]	Isolation Forest labels readings; Random Forest and Decision Tree predict anomaly occurrence from label sequences.	Shows automatic labeling can feed supervised prediction; the two supervised models perform similarly.	Predicts anomalies rather than theft specifically, and labels come from an unsupervised model without ground truth.
Short-Term Load Forecasting in Power Systems Based on the Prophet-BO-XGBoost Model [6]	XGBoost for nonlinear feature-load relations, Prophet for time-series label prediction, Bayesian optimization for tuning.	Targets the accuracy and generalization limits of short-term load forecasting under complex nonlinear conditions.	Built on grid-level load data; behavior on single-household loads from a low-cost meter is not established.

Research Paper Title	Focus and Method	Main Contribution / Findings	Limitations / Interpretation
Real Time Energy Monitoring & Current Spikes Detection System for Enhanced Energy Conservation in Residential Area [7]	ESP32, ZMPT101B and SCT013-030 prototype with ThingSpeak cloud reporting and bill estimation.	Achieves real-time monitoring at low cost with a highest observed error difference of 12.438%.	Monitoring only: no forecasting, theft classification or load cutoff. Error level calls for calibration.

CHAPTER 5: EVALUATION STRATEGY

A research-grade theft-aware meter should be evaluated at three levels: the sensor, the models, and the full system. The reviewed studies report detection rate and false positive rate [1], accuracy, sensitivity and error counts [2], and accuracy with AUC [3]. The hardware study reports measurement error [7].

5.1 CLASSIFICATION PERFORMANCE

Precision, Recall, F1, AUC, detection rate, and false positive rate should all be reported because accuracy alone hides poor recall on the rare theft class [1,2,3]. Because VoltIQ can trigger a relay, the false positive rate matters most: a wrongly disconnected honest consumer is a real harm. Per-consumer-type results should also be reported, since performance differs sharply between commercial and residential profiles [3].

5.2 FORECAST ACCURACY

Forecast error should be measured on held-out time periods with standard error measures such as MAE and RMSE, and compared against simple baselines. The Prophet-BO-XGBoost study is motivated by accuracy and generalization [6], so VoltIQ should test generalization to unseen days and loads rather than only in-sample fit.

5.3 ANOMALY DETECTION QUALITY

Field data rarely come with anomaly labels, and Isolation Forest is itself used to create labels in [5]. Evaluation should therefore combine injected anomalies, such as simulated bypass or tampering events, with review of flagged windows. Agreement-based alerting, as in [4], should be compared against a single detector.

5.4 MEASUREMENT AND SYSTEM-LEVEL EVALUATION

Sensor readings should be compared with a reference meter and reported as percentage error, the measure used in [7]. System-level tests should record relay response time, false-trip rate, logging continuity during connectivity loss, and dashboard update delay. These design checks are proposed here and are not reported by the reviewed studies.

CHAPTER 6: REPORTED FINDINGS FROM THE SUPPLIED LITERATURE

First, the XGBoost theft detector in [1] reports a 96% detection rate and a 3% false positive rate, and the XGBoost with SMOTE study [2] reports 93.63% accuracy. This supports XGBoost as a strong classifier for theft detection.

Second, the multi-model comparison in [3] finds accuracy above 95% and AUC near 0.99 for the best tree ensemble, and shows that consumer-specific models outperform global models, with residential profiles remaining the hardest case.

Third, the hybrid anomaly detector in [4] outperforms single-method models by requiring agreement among four detectors, and [5] shows that Isolation Forest labels can drive supervised anomaly prediction. The hardware prototype in [7] reports a highest error difference of 12.438%.

These results should be interpreted within each paper's dataset, preprocessing, and evaluation protocol. They are evidence for specific study settings, not a head-to-head benchmark. A fair comparison would require a shared dataset, identical splits, and the same metrics.

CHAPTER 7: CHALLENGES AND RESEARCH DIRECTIONS

7.1 IMBALANCE AND SYNTHETIC THEFT DATA

Real theft examples are scarce. Artificially created attacks [1] and SMOTE [2,3] help, but synthetic patterns may not match real tampering. A field trial that records labeled bypass tests under supervision would give VoltIQ more realistic data.

7.2 GENERALIZATION ACROSS CONSUMER TYPES

Models that work well for commercial users can struggle on residential profiles [3]. VoltIQ should state its target consumer type clearly and test across household sizes and appliance mixes.

7.3 FALSE POSITIVES AND SAFE LOAD CUTOFF

Low false positive rates are a stated strength of the reviewed classifiers [1,2], and agreement-based detection further reduces false alarms [4]. Even so, cutting power on a model decision carries risk. Persistence checks, alert-before-cut stages, and a manual override are advisable design safeguards.

7.4 SENSOR ACCURACY AND CALIBRATION

Low-cost sensors can show errors of more than 12% [7]. Calibration, temperature effects, and sampling rate directly limit what the ML layers can detect.

7.5 EDGE CONSTRAINTS AND CONNECTIVITY

The ESP32 is suited to sensing and logging, while heavier models such as XGBoost, Random Forest, and Prophet can run on a server or laptop. SD buffering [Section 3.1] keeps data safe during outages, but the delay between measurement and alert must be measured.

7.6 PRIVACY AND RESPONSIBLE USE

Fine-grained consumption data can reveal household behavior. Deployments should limit data retention, restrict access, and explain to consumers why a disconnection occurred.

CHAPTER 8: DISCUSSION

The literature suggests that the most promising systems are compositional. Tree-ensemble classifiers provide accurate theft decisions [1,2,3], unsupervised detectors supply labels and early warnings [4,5], and hybrid forecasting provides an expected-usage baseline [6]. Low-cost hardware makes the whole stack affordable [7].

The main gap is integration. The hardware study stops at monitoring [7], the anomaly study works on simulated data [4], and none of the reviewed work closes the loop with an automatic relay. VoltIQ's contribution is therefore the pipeline as a whole: a single device and software stack that measures, forecasts, flags, classifies, and acts. Its modular design also allows each component to be measured independently and replaced without redesigning the system.

CHAPTER 9: LIMITATIONS OF THIS REVIEW

This paper is a reference-guided synthesis of seven studies and the VoltIQ project plan. It does not report a new dataset, trained model, field trial, or statistically controlled experiment, and VoltIQ has not yet been validated. The review draws on the published abstracts and summaries of the cited works, so claims should be checked against the full texts before submission. A broader systematic review would require a larger search strategy, inclusion criteria, and coverage of additional studies, including work on relay-based load control.

CHAPTER 10: CONCLUSION

Electricity monitoring has progressed from low-cost sensing and billing estimation [7], to supervised theft classification [1,2,3], to unsupervised anomaly detection and dashboards [4,5], and to hybrid load forecasting [6]. Each stage addresses a weakness of the previous one, but no reviewed study connects all of them.

The resulting design principle is a layered system in which sensing, logging, forecasting, anomaly detection, theft classification, and relay control work together under conservative confirmation rules. Future work should evaluate such a system not only on accuracy but also on false positive rate, measurement error, latency, and safety. VoltIQ's planned testing and field trial are intended to provide that evidence.

CHAPTER 11: REFERENCES

[1] A. I. Kawoosa, D. Prashar, M. Faheem, N. Jha, and A. A. Khan, "Using machine learning ensemble method for detection of energy theft in smart meters," IET Generation, Transmission & Distribution, vol. 17, no. 21, 2023. doi: 10.1049/gtd2.12997.

[2] R. I. Putro, G. D. Putra, and Y. S. Wijoyo, "Electricity Theft Detection Based on Load Profile Using XGBoost and SMOTE," 2025, pp. 126-130. [Add venue/proceedings name.]

[3] F. Alfaverh, H. Gan, V. Miroshnyk, Z. Bin Saeed, I. Blinov, P. Shymaniuk, P. Tarassodi, and I. Mporas, "Electricity Theft Detection from Electricity, Gas and Water Measurements using Machine Learning," 2026. [Add venue; available via the University of Hertfordshire repository.]

[4] [Authors to be added], "Empowering energy management: anomaly detection in smart meter data for proactive consumption control," Bulletin of Electrical Engineering and Informatics. [Add volume, year and DOI from beei.org.]

[5] R. ELhadad, Y. F. Tan, and W. N. Tan, "Anomaly Prediction in Electricity Consumption Using a Combination of Machine Learning Techniques," International Journal of Technology, vol. 13, no. 6, p. 1317, 2022.

[6] S. Zeng, C. Liu, H. Zhang, B. Zhang, and Y. Zhao, "Short-Term Load Forecasting in Power Systems Based on the Prophet-BO-XGBoost Model," Energies, vol. 18, no. 2, p. 227, 2025.

[7] [Authors to be added], "Real Time Energy Monitoring & Current Spikes Detection System For Enhanced Energy Conservation In Residential Area," UTHM periodical (PEAT). [Add volume, year and DOI.]